

M1: the local number bound without space-averaging

Abstract

Milestone M0 needs norm-compactness of $\{\omega_g|_{\mathfrak{A}(O)}\}$, i.e. the hypothesis $\omega(N_{\xi,\tau}) \leq M(\xi)$ of [2, Thm. 4.1], for the *unaveraged* states ω_g . We settle this, with one correction to the plan.

(1) The strategy’s proposed route does not work. We display Glimm and Jaffe’s proof of Theorem 3.3.1(a) explicitly. Its only input is the *extensive* bound $\omega_{g_n}(N) = O(n)$ coming from Theorem 5.1 ($-M \leq E_g \leq 0$, $M \propto \text{volume}$); the averaging, performed at the scale n of the cutoff, contributes exactly the compensating factor $1/n$. Averaging at any scale R independent of the volume yields $O(V/R)$ and is useless. So the averaging is not a technical convenience: it is the device converting an extensive bound into an intensive one, and it cannot be removed by bookkeeping. In particular the route proposed in the strategy note — “GJ IV Theorem A_g plus the linear lower bound” — fails, because the linear lower bound *is* Theorem 5.1, i.e. the extensive input itself.

(2) What is true, and suffices. A Chebyshev selection applied to Glimm and Jaffe’s own averaged estimate produces a *cofinal sequence* of cutoffs along which the unaveraged bound holds, simultaneously for a countable family of test functions (Theorem 3.3). This is exactly the hypothesis M0 requires, so:

there is a locally normal ground state of $(\mathfrak{A}, \alpha)$ arising as a norm-local limit of *unaveraged* finite-volume ground states; and under uniqueness the limit is the vacuum (Corollaries 4.1–4.2).

(3) The residual gap, identified. Convergence along the *full net* needs $\sup_{g \in \mathcal{G}} \omega_g(N_{\xi,\tau}) < \infty$, an *intensive* bound on the local particle density. We show in §5 that this is a Hamiltonian form of Problem 1 of [4]. Hence M0 removes Problem 1 from the *identification* step, but a Problem-1-like statement survives in the *compactness* step unless one accepts the cofinal-sequence version.

1 What is needed

We use the notation of [1]. Let $N_{\xi,\tau}$ denote the local number operator of [2, §3.2]: the Fock quantisation of the one-particle operator built from $\xi \geq 0$ and the energy weight μ^τ . We use only three of its properties, all explicit in [2, §3.2–3.3]:

- (N1) $N_{\xi,\tau} \geq 0$ as a self-adjoint operator, so $\omega(N_{\xi,\tau}) \in [0, \infty]$ for every state ω ;
- (N2) $U_a N_{\xi,\tau} U_a^* = N_{\xi(\cdot - a), \tau}$, U_a the space translation;
- (N3) the constant M in Theorem 3.3.1 is *translation invariant*: $M(\xi(\cdot - a)) = M(\xi)$.

Theorem 1.1 (Glimm–Jaffe [2, Thm. 4.1]). *Let (ω_n) be a sequence of normal states on $\mathfrak{A}$ and suppose that for some $\tau > 0$ and each $0 \leq \xi \in C_0^\infty$,*

$$\omega_n(N_{\xi,\tau}) \leq M(\xi) \tag{1}$$

with $M(\xi)$ independent of n . Then $(\omega_n|_{\mathfrak{A}(B)})$ lies in a norm-compact subset of $\mathfrak{A}(B)^$ and every limit point is normal.*

Theorem 1.1 is a statement about *arbitrary* sequences of normal states. So M1 is exactly: produce a cofinal family of g 's for which (1) holds with ω_n replaced by ω_g .

2 Why the averaging cannot simply be deleted

Glimm and Jaffe verify (1) for the averaged states

$$\omega_n(A) = \frac{1}{n} \int \langle \Omega_{g_n}, \sigma_\alpha(A) \Omega_{g_n} \rangle h(\alpha/n) d\alpha = \int_{-1}^1 h(u) \omega_{g_n}(\tau_{nu} A) du, \quad (2)$$

where $g_n(x) = g(x/n)$ with $g \equiv 1$ on $[-3, 3]$, and $h \geq 0$, $\text{supp } h \subseteq [-1, 1]$, $\int h = 1$ [2, (2.3)–(2.5)].

Proposition 2.1 (the mechanism). *The proof of [2, Thm. 3.3.1(a)] is the chain*

$$\begin{aligned} \omega_n(N_B) &= \langle \Omega_{g_n}, C_n \Omega_{g_n} \rangle, \quad \|(N + I)^{-1/2} C_n (N + I)^{-1/2}\| \leq \text{const} \cdot \frac{1}{n} |B|, \\ \implies \omega_n(N_B) &\leq \text{const} \cdot \frac{1}{n} |B| \omega_{g_n}(N) \leq \text{const} \cdot |B|, \end{aligned}$$

the last step by [2, Thm. 5.1]. The operator norm bound comes from

$$\sup_{x,y} \left| \frac{1}{n} \int E(x - \alpha) E(y - \alpha) h(\alpha/n) d\alpha \right| \leq \frac{1}{n} \|h\|_\infty \|E\|_1 = \frac{1}{n} \|h\|_\infty |B|.$$

Two consequences, both negative.

Remark 2.2 (the input is extensive). [2, Thm. 5.1] states $-M \leq E_g \leq 0$ with M proportional to the volume of $(\text{supp } g)_1$, and its Remark 2 draws from this precisely that the expected number of bare particles *per unit volume* is finite. Thus the only available bound on the total number is $\omega_{g_n}(N) = O(n)$, and it is sharp: the ground state contains $O(\text{volume})$ particles.

Remark 2.3 (the averaging scale must match the volume). Replacing $h(\alpha/n)/n$ in (2) by $h(\alpha/R)/R$ with R fixed changes the operator norm bound to $\text{const} \cdot |B|/R$ and hence gives $\omega_g^{(R)}(N_B) \leq \text{const} \cdot |B| V(g)/R$, which is uniform in g only if $R \asymp V(g)$. So no fixed-scale averaging, and *a fortiori* no deletion of the averaging, survives this argument.

Remark 2.4 (correction to the strategy). The strategy note proposed deriving the unaveraged bound from “GJ IV Theorem A_g plus the linear lower bound $-MV \leq E(g)$ ”. This cannot work: the linear lower bound is Theorem 5.1, i.e. the extensive input of Remark 2.2, and Theorem A_g of [3] bounds *field* perturbations, not local particle number. The proposed route is therefore withdrawn.

3 The selection theorem

Lemma 3.1 (the average is an average of unaveraged states). *For every $A \in \mathfrak{A}$ and every positive quadratic form Q on the Fock space,*

$$\omega_n(A) = \int_{-1}^1 h(u) \omega_{\tau_{-nu} g_n}(A) du, \quad \omega_n(Q) = \int_{-1}^1 h(u) \omega_{\tau_{-nu} g_n}(Q) du,$$

where $\omega_g(Q) := Q[\Omega_g] \in [0, \infty]$.

Proof. Substituting $\alpha = nu$ in (2) gives $\omega_n(A) = \int h(u) \omega_{g_n}(\tau_{nu}A) du$. By [1, Lem. 3.2], $\omega_{\tau ag} = \omega_g \circ \tau_{-a}$, hence $\omega_{g_n} \circ \tau_{nu} = \omega_{\tau_{-nu}g_n}$. The statement for Q follows from the same identity at the level of quadratic forms, using (N2) and $\Omega_{\tau_{-nu}g_n} = U_{nu}^* \Omega_{g_n}$ up to a phase. $\square$

Lemma 3.2 (measurability). *For fixed n and ξ , the map $u \mapsto \omega_{\tau_{-nu}g_n}(N_{\xi,\tau}) \in [0, \infty]$ is lower semicontinuous, hence Borel measurable.*

Proof. Write $F(u) = q[U_{nu}^* \Omega_{g_n}]$ where $q[\psi] = \|N_{\xi,\tau}^{1/2} \psi\|^2$ is the closed positive quadratic form of $N_{\xi,\tau}$. Closed positive forms are lower semicontinuous on the Hilbert space, and $u \mapsto U_{nu}^* \Omega_{g_n}$ is norm-continuous by strong continuity of U . The composition of a lower semicontinuous function with a continuous map is lower semicontinuous. $\square$

Theorem 3.3 (M1). *Fix $\tau \in (0, 1)$ and a countable family $\{\xi_k\}_{k \geq 1} \subset C_0^\infty(\mathbb{R})$ with $\xi_k \geq 0$. Let $M_k := \max\{M(\xi_k), 1\}$ with $M(\xi_k)$ the constant of [2, Thm. 3.3.1(b)]. Then there are $u_n \in [-1, 1]$ such that the cutoffs*

$$\tilde{g}_n := \tau_{-nu_n} g_n \in \mathcal{G}$$

satisfy

- (i) $\tilde{g}_n \equiv 1$ on $[-2n, 2n]$; in particular $d(\tilde{g}_n, O) \rightarrow \infty$ for every bounded O , so $(\tilde{g}_n)$ is cofinal in $\mathcal{G}$;
- (ii) $\omega_{\tilde{g}_n}(N_{\xi_k,\tau}) \leq 2^{k+1} M_k$ for all $n \geq 1$ and all $k \geq 1$.

Proof. Fix n . Let $\nu(du) := h(u) du$, a Borel probability measure on $[-1, 1]$, and put $F_k(u) := \omega_{\tau_{-nu}g_n}(N_{\xi_k,\tau}) \geq 0$, measurable by Lemma 3.2. By Lemma 3.1 and [2, Thm. 3.3.1(b)],

$$\int F_k d\nu = \omega_n(N_{\xi_k,\tau}) \leq M(\xi_k) \leq M_k.$$

Set $S_k := \{u : F_k(u) > 2^{k+1} M_k\}$. By Chebyshev's inequality,

$$\nu(S_k) \leq \frac{1}{2^{k+1} M_k} \int F_k d\nu \leq 2^{-(k+1)},$$

whence $\nu(\bigcup_{k \geq 1} S_k) \leq \sum_{k \geq 1} 2^{-(k+1)} = \frac{1}{2} < 1 = \nu([-1, 1])$. Therefore $[-1, 1] \setminus \bigcup_k S_k$ has ν -measure at least $\frac{1}{2}$, in particular it is nonempty; choose u_n in it. Then $F_k(u_n) \leq 2^{k+1} M_k$ for every k , which is (ii).

For (i): $(\tau_{-a}g)(x) = g(x+a)$, so $\tilde{g}_n(x) = g_n(x+nu_n)$ equals 1 exactly when $x+nu_n \in \{g_n = 1\} \supseteq [-3n, 3n]$, i.e. on $[-3n-nu_n, 3n-nu_n] \supseteq [-2n, 2n]$ because $|u_n| \leq 1$. Hence for $O \subseteq [-R, R]$ we get $d(\tilde{g}_n, O) \geq 2n - R \rightarrow \infty$. $\square$

Remark 3.4. The constants $2^{k+1} M_k$ grow in k . This is harmless: Theorem 1.1 requires, for each fixed ξ , a constant independent of n , which (ii) provides.

4 Consequences

Corollary 4.1 (existence). *Let $\{\xi_k\}$ be chosen so that for every bounded open O there is k with $\xi_k \geq 1$ on O . Then the sequence $(\omega_{\tilde{g}_n})$ of Theorem 3.3 satisfies the hypothesis (1) of Theorem 1.1. Consequently it admits a subsequence converging in norm on every $\mathfrak{A}(O)$ to a locally normal state ω , and by [1, Thm. 4.2] ω is a ground state of $(\mathfrak{A}, \alpha)$. By [1, Lem. 3.3] ω is $\mathbb{Z}_2$ -invariant when P is even.*

Proof. Combine Theorem 3.3(ii) with Theorem 1.1, then run the diagonal argument of [1, Cor. 4.3] over an exhausting sequence of regions; Theorem 3.3(i) supplies the hypothesis $d(\tilde{g}_n, O) \rightarrow \infty$ of [1, Thm. 4.2]. $\square$

Corollary 4.2 (convergence, under uniqueness). *If $(\mathfrak{A}, \alpha)$ has a unique locally normal, $\mathbb{Z}_2$ - and translation-invariant ground state ω_∞ , then $\omega_{\tilde{g}_n} \rightarrow \omega_\infty$ in norm on every $\mathfrak{A}(O)$ along the full sequence $(\tilde{g}_n)$.*

Proof. Every subsequence has, by Corollary 4.1, a further subsequence converging in norm on each $\mathfrak{A}(O)$ to a locally normal ground state, which is ω_∞ by hypothesis. A sequence in a metric space all of whose subsequences have sub-subsequences converging to the same limit converges to that limit. $\square$

5 The residual gap, precisely

Problem 5.1 ((H2)-full). Show that for some $\tau > 0$ and each $0 \leq \xi \in C_0^\infty$,

$$\sup_{g \in \mathcal{G}} \omega_g(N_{\xi, \tau}) < \infty.$$

Problem 5.1 is what upgrades Corollary 4.2 from a cofinal sequence to the full net, i.e. to Theorem A as stated in the strategy note. It asks for an *intensive* bound: the expected local particle density at a *given* location, uniformly in the cutoff, whereas Theorem 5.1 of [2] delivers only the density *averaged over the volume*.

Three remarks on how to attack it, and one on why it is not cheap.

1. **A naive operator inequality is unavailable.** One would like $N_{\xi, \tau} \leq c_1 H(g) + c_2(\xi)$ with c_2 independent of g , since $H(g)\Omega_g = 0$ would then give the bound at once. But the only route to comparing H_0 with $H(g)$ is $-H_I(g) \leq \varepsilon H_0 + b(g)$, giving $(1 - \varepsilon)H_0 \leq H(g) + b(g) + E(g)$, in which $b(g)$ is extensive while Theorem 5.1 bounds $E(g)$ only from *below*, $-MV \leq E_g$. The volume terms do not cancel.
2. **A homogeneity estimate would suffice.** By [1, Lem. 3.2], $a \mapsto \omega_g(N_{\xi(\cdot - a), \tau}) = \omega_{\tau - a, g}(N_{\xi, \tau})$, and Theorem 3.3 bounds its ν -average. Any Harnack-type statement — that this function varies by at most a constant factor as a ranges over the bulk of $\{g = 1\}$ — converts the averaged bound into the pointwise one.
3. **It is a Hamiltonian form of GRS Problem 1.** $\omega_g(N_{\xi, \tau})$ is, up to the energy weight, a smeared $\langle : \varphi^2 : \rangle$ -type local density in the finite-volume vacuum; its uniform boundedness in the volume is exactly the kind of local L^p control of the vacuum that [4, Problem 1] conjectures in the form $\log \|\nu_{\Lambda \Lambda'}\|_p = O(|\Lambda|)$, and which they observe would re-prove the locally Fock property.

Remark 5.2 (what this does to the strategy’s headline). The strategy claims M0 replaces GRS Problem 1 by an exact identity. That is correct for the *identification* step, and it is the substantive gain. But §2 shows a Problem-1-like statement survives in the *compactness* step. Theorem 3.3 buys it off at the price of restricting to a cofinal sequence. The honest formulation of the programme’s current reach is therefore:

modulo uniqueness, the finite-volume ground states converge along a cofinal sequence of spatial cutoffs, in norm on every local algebra, to the vacuum — with the full-net statement contingent on Problem 5.1.

This is still the Hamiltonian analogue of what Nelson and Guerra–Rosen–Simon obtain Euclidean-ly, and it is what M2 verifies is correct in the solvable model.

References

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